

Basic Motion Directions of Motorized Platform for Applied Motor Voltages

This platform moves the pen in a two-dimensional Cartesian plane by using two DC motors.

The motion relation between motors and pen is described in Table 1 and Figure 1 of this section.

First mentioning abbreviations used in this section.

- H: Pen moves horizontal
- V: Pen moves vertical
- R: Pen moves right (in horizontal direction)
- L: Pen moves left (in horizontal direction)
- U: Pen moves up (in vertical direction)
- D: Pen moves down (in vertical direction)
- Dg: Pen moves diagonal
- +: Motor turns RIGHT (clockwise, CL)
- -: Motor turns LEFT (counter clockwise, CC)
- 0: Motor stop

Direction of Pen	Mot-M1	Mot-M2
H-R	+	+
H-L	-	-
V-U	-	+
V-D	+	-
Dg-RU	0	+
Dg-RD	+	0
Dg-LU	-	0
Dg-LD	0	-

Table 1: Motion relation between motors direction and pen moving in motorized platform

In Figure 1 the relation is shown graphically on the platform area.

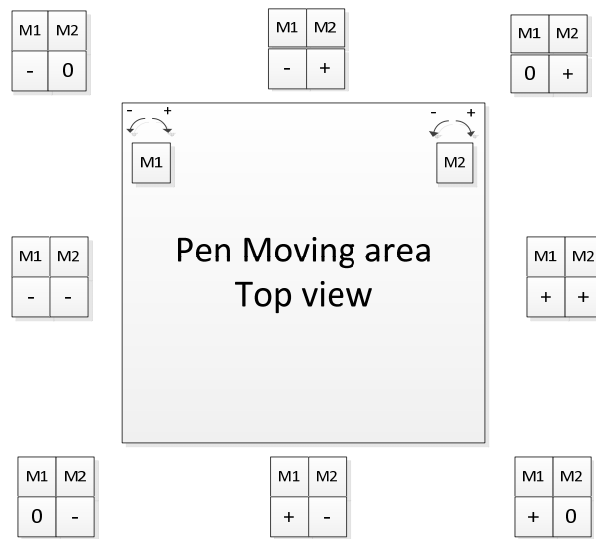


Figure 1: Motion relation between motors direction and pen moving in motorized platform

In Figure 2 the placement of limit switches on the motorized platform are shown.

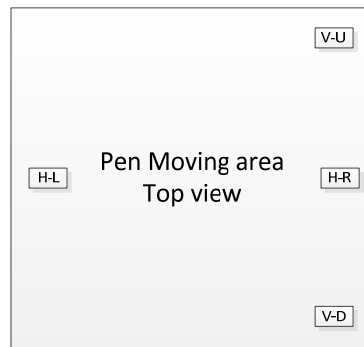


Figure 2: Four limit switches on the motorized platform